Safety Data Sheet (SDS)

1. Substance/preparation and company identification

Ares SN Herbicide

Major Recommended Uses:
Use: crop protection product, herbicide

Company:
BASF Paraguaya S.A.
Avda. Santa Teresa 2106 e/ Avda. Aviadores del Chaco y Herminio Maldonado
Edificio “Torres del Paseo” Torre II - Piso 11 - Of. 6 -
CP 1821 Asuncion, PARAGUAY
Telephone: +595 21 695641 / 695642 / 695643
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Centro de Toxicología: +595 21 220 418
Telephone: 0098005410053/+55 12 3128-1590

2. Hazards Identification

Classification of the substance or mixture

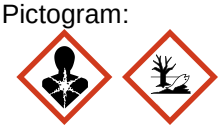
According to UN GHS criteria

Hazardous to the aquatic environment - acute: Cat. 1
Reproductive toxicity: Cat. 2 (unborn child)
Hazardous to the aquatic environment - chronic: Cat. 1

Label elements

According to UN GHS criteria

Pictogram:



Signal Word:
Warning

Hazard Statement:

H361	Suspected of damaging the unborn child.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

Precautionary Statement:

P101	If medical advice is needed, have product container or label at hand.
P102	Keep out of reach of children.
P103	Read carefully and follow all instructions.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P201	Obtain special instructions before use.
P202	Do not handle until all safety precautions have been read and understood.

Precautionary Statements (Response):

P391	Collect spillage.
P308 + P313	IF exposed or concerned: Get medical attention.

Precautionary Statements (Storage):

P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

According to UN GHS criteria

Labelling of special preparations:
May produce an allergic reaction. Contains: 1,2-benzisothiazol-3(2H)-one

Other hazards

According to UN GHS criteria

Other Hazards (GHS):

See section 12 - Results of PBT and vPvB assessment.

Other Hazards (GHS):

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

Assessment PBT / vPvB:
The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria..

3. Composition/Information on Ingredients

Mixtures

Chemical nature

crop protection product, herbicide, Soluble concentrate (SL)

Hazardous ingredients (GHS)
According to UN GHS criteria

imazamox

Content (W/W): 3,12 % CAS Number: 114311-32-9	Reproductive toxicity: Cat. 2 (unborn child) Hazardous to the aquatic environment - acute: Cat. 1 Hazardous to the aquatic environment - chronic: Cat. 1 M-factor acute: 10 M-factor chronic: 10 H361, H400, H410

imazapyr (ISO); 2-[4,5-dihydro-4-methyl-4-(1-methylethyl)-5-oxo-1H-imidazol-2-yl]-3-pyridine carboxylate

Content (W/W): 1,38 % CAS Number: 81334-34-1 INDEX-Number: 613-126-00-1	Serious eye damage/eye irritation: Cat. 2B Hazardous to the aquatic environment - acute: Cat. 3 H320, H402

Ammonium hydroxide ((NH4)(OH))

Content (W/W): < 1 % CAS Number: 1336-21-6	Corrosive to metals: Cat. 1 Acute toxicity: Cat. 4 (oral) Skin corrosion/irritation: Cat. 1B Serious eye damage/eye irritation: Cat. 1 Specific target organ toxicity — single exposure: Cat. 3 (irr. to respiratory syst.) Hazardous to the aquatic environment - acute: Cat. 1 Hazardous to the aquatic environment - chronic: Cat. 2 H290, H302, H335, H314, H411, H400

1,2-benzisothiazol-3(2H)-one

Content (W/W): < 0,05 % CAS Number: 2634-33-5 EC-Number: 220-120-9	Acute toxicity: Cat. 4 (oral) Skin corrosion/irritation: Cat. 2 Serious eye damage/eye irritation: Cat. 1 Skin sensitization: Cat. 1 Hazardous to the aquatic environment - acute: Cat. 1 Hazardous to the aquatic environment - chronic: Cat. 1 M-factor acute: 1 M-factor chronic: 1 H318, H315, H302, H317, H400, H410

propane-1,2-diol
Content (W/W): < 15 %
CAS Number: 57-55-6
EC-Number: 200-338-0

For hazard statements not written out in full in this section, the full text is listed in section 16.

4. First-aid measures

General advice:
Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-fighting measures

Suitable extinguishing media:
water spray, carbon dioxide, foam, dry powder

Specific hazards:
carbon monoxide, carbon dioxide, nitrogen oxides, sulfur oxides, Phosphorus compounds, ammonia oxides
The substances/groups of substances mentioned can be released in case of fire.

Further information:
In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

6. Accidental release measures

Personal precautions, protective equipment and emergency measures

Personal precautions:
Use personal protective clothing. Avoid contact with the skin, eyes and clothing. Do not breathe vapour/spray.

Environmental precautions:
Do not discharge into the subsoil/soil. Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations.

7. Handling and storage

Handling
Technical measures:
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended.

Fire and explosion protection:
No special precautions necessary. The substance/product is non-combustible. Product is not explosive.

Precautions/ Advice on safe handling:
No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Specific hygiene measures:

Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

Storage

Technical measures:

Protect from temperatures below: -10 °C
The product can crystallize below the limit temperature.
Protect from temperatures above: 40 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Incompatible products and materials:
Segregate from foods and animal feeds.

8. Exposure controls and personal protection

Specific control parameters

Components with occupational exposure limits:

No substance specific occupational exposure limits known.

Personal protective equipment
Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Skin and body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

9. Physical and chemical properties

State of matter:	liquid (20 °C)
Form:	liquid
Colour:	yellow to amber
Odour:	aliphatic

pH value: approx. 5 - 7
(approx. 20 °C)
(measured with the undiluted substance)

Specific temperatures or temperature ranges at which changes in physical state occur

solidification temperature: approx. -14 °C
(1.013,3 hPa)

Boiling point: approx. 100 °C
(1.013,3 hPa)
Information applies to the solvent.

Flash point:
Non-flammable.

Lower explosion limit:
As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.

Upper explosion limit:
As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.

Thermal decomposition: No decomposition if stored and handled as prescribed/indicated.

Explosion hazard: Based on the chemical structure there is no indication of explosive properties.

Fire promoting properties: Based on its structural properties the product is not classified as oxidizing.

Vapour pressure: approx. 23,4 hPa
(20 °C)
Information applies to the solvent.

VOC content: No data available.

Relative vapour density (air):
not applicable

Density: approx. 1,08 g/cm3
(20 °C)

Relative density: No data available.

Solubility in water: soluble

Partitioning coefficient n-octanol/water (log Pow):
not applicable for mixtures

Self-ignition temperature: 398 °C

Self ignition: No data available.

Odour threshold limit value:
Not determined due to potential health hazard by inhalation.

Evaporation rate:
not applicable

Flammability: not applicable

Viscosity, dynamic:	approx. 83 mPa.s (20 °C)	(calculated (from kinematic viscosity))

Other information:
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

10. Stability and reactivity

Reactivity:
No hazardous reactions if stored and handled as prescribed/indicated.

Chemical stability:
The product is stable if stored and handled as prescribed/indicated.

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Conditions to avoid:
See SDS section 7 - Handling and storage.

Incompatible materials and substances:
strong bases, strong acids, strong oxidizing agents

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological information

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact. Virtually nontoxic by inhalation. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

LD50 rat, male/female(oral): > 5.000 mg/kg (OECD Guideline 401)

LC50 rat, male/female (by inhalation): > 6,18 mg/l 4 h (OECD Guideline 403)

LD50 rabbit, male/female (dermal): > 5.000 mg/kg (OECD Guideline 402)

Local effects

Assessment of irritating effects:
Not irritating to the eyes. Not irritating to the skin. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Primary skin irritation rabbit: non-irritant

Eye irritation rabbit: non-irritant (OECD Guideline 405)

Assessment other acute effects

Assessment other acute effects:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Sensitization

Assessment of sensitization:
There is no evidence of a skin-sensitizing potential.The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Buehler test guinea pig: Non-sensitizing. (OECD Guideline 406)

Genetic toxicity

Assessment of mutagenicity:
Mutagenicity tests revealed no genotoxic potential. The product has not been tested. The statement has been derived from the properties of the individual components.

Carcinogenicity

Assessment of carcinogenicity:
The results of various animal studies gave no indication of a carcinogenic effect. The product has not been tested. The statement has been derived from the properties of the individual components.

Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from the properties of the individual components.

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: imazamox

Assessment of teratogenicity:
Indications of possible developmental toxicity/teratogenicity were seen in animal studies.

Repeated dose toxicity

Assessment of repeated dose toxicity:
No substance-specific organtoxicity was observed after repeated administration to animals. The product has not been tested. The statement has been derived from the properties of the individual components.

Aspiration Hazard

Assessment of Aspiration Hazard:
No aspiration hazard expected., The product has not been tested. The statement has been derived from the properties of the individual components.

Other relevant toxicity information

Misuse can be harmful to health.

12. Ecological information

Possible environmental effects, behaviour and fate

Ecotoxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: imazapyr (ISO); 2-[4,5-dihydro-4-methyl-4-(1-methylethyl)-5-oxo-1H-imidazol-2-yl]-3-pyridine carboxylate
Toxicity to fish:
LC50 (96 h) > 100 mg/l, Oncorhynchus mykiss

Information on: imazamox
Toxicity to fish:
LC50 (96 h) > 122 mg/l, Oncorhynchus mykiss

Information on: imazapyr (ISO); 2-[4,5-dihydro-4-methyl-4-(1-methylethyl)-5-oxo-1H-imidazol-2-yl]-3-pyridine carboxylate
Aquatic invertebrates:
EC50 (48 h) > 100 mg/l, Daphnia magna

Information on: imazamox
Aquatic invertebrates:
EC50 (48 h) > 100 mg/l, Daphnia magna

Information on: imazapyr (ISO); 2-[4,5-dihydro-4-methyl-4-(1-methylethyl)-5-oxo-1H-imidazol-2-yl]-3-pyridine carboxylate
Aquatic plants:
EC50 (7 d) 11,7 mg/l, Anabaena flos-aquae

No observed effect concentration 5,26 mg/l, Anabaena flos-aquae

Information on: imazamox
Aquatic plants:

EC50 (72 h) 29,1 mg/l (growth rate), Pseudokirchneriella subcapitata

EC50 (7 d) 0,0085 mg/l (growth rate), Lemna gibba

EC10 (7 d) 0,0036 mg/l (growth rate), Lemna gibba

Information on: imazapyr (ISO); 2-[4,5-dihydro-4-methyl-4-(1-methylethyl)-5-oxo-1H-imidazol-2-yl]-3-pyridine carboxylate
Chronic toxicity to fish:
No observed effect concentration (33 d) 118 mg/l, Pimephales promelas

Information on: imazamox
Chronic toxicity to fish:
No observed effect concentration (28 d) 1,22 mg/l, Cyprinodon variegatus
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Information on: imazapyr (ISO); 2-[4,5-dihydro-4-methyl-4-(1-methylethyl)-5-oxo-1H-imidazol-2-yl]-3-pyridine carboxylate
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d), 97,1 mg/l, Daphnia magna

Information on: imazamox
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d), > 10 mg/l, Daphnia magna (semistatic)

Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: imazapyr (ISO); 2-[4,5-dihydro-4-methyl-4-(1-methylethyl)-5-oxo-1H-imidazol-2-yl]-3-pyridine carboxylate
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Information on: imazamox
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Bioaccumulation

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: imazamox
Bioaccumulation potential:
Bioconcentration factor: < 1, Lepomis macrochirus (OECD Guideline 305)
Does not accumulate in organisms.

Information on: imazapyr (ISO); 2-[4,5-dihydro-4-methyl-4-(1-methylethyl)-5-oxo-1H-imidazol-2-yl]-3-pyridine carboxylate
Bioaccumulation potential:
Bioconcentration factor: < 1,0, Lepomis macrochirus
Does not accumulate in organisms.

Mobility

Assessment transport between environmental compartments:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: imazamox
Assessment transport between environmental compartments:
The substance will not evaporate into the atmosphere from the water surface.
Following exposure to soil, the product trickles away and can - dependant on degradation - be transported to deeper soil areas with larger water loads.

Information on: imazapyr (ISO); 2-[4,5-dihydro-4-methyl-4-(1-methylethyl)-5-oxo-1H-imidazol-2-yl]-3-pyridine carboxylate
Assessment transport between environmental compartments:
The substance will not evaporate into the atmosphere from the water surface.
Following exposure to soil, the product trickles away and can - dependant on degradation - be transported to deeper soil areas with larger water loads.

Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

13. Disposal considerations

Methods for safe and environmentally preferred disposal
Product: Must be sent to a suitable incineration plant, observing local regulations.

Product residues: Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport information

Land transport

Hazard Class:	9
Packing group:	III

UN Number:	3082
Hazard label:	9
Risk Number:	90
Proper Shipping Name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (IMAZAMOX)

Waterway Transport

IMDG

Hazard class:	9
Packing group:	III
UN Number:	3082
Hazard label:	9, EHSM
Marine pollutant:	YES
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (IMAZAMOX)

Air transport

IATA/ICAO

Hazard class:	9
Packing group:	III
UN Number:	3082
Hazard label:	9, EHSM
Proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (IMAZAMOX)

Further information

Product may be shipped as non-hazardous in suitable packages containing a net quantity of 5 L or less under the provisions of various regulatory agencies: ADR, RID, ADN: Special Provision 375; IMDG: 2.10.2.7; IATA: A197; TDG: Special Provision 99(2); 49CFR: §171.4 (c) (2) and also the Special Provision 375 in Appendix B which is regulated in China "Regulations Concerning Road Transportation of Dangerous Goods Part 3: Index of dangerous goods name and transportation requirements" (JT/T 617.3)

Additional Information

Classification of land transport generated according to the criteria of Decree 17723:1997.

15. Regulatory information

Other regulations

To avoid risks to man and the environment, comply with the instructions for use.

16. Other information

Full text of hazard statements, if mentioned in section 3:

H361	Suspected of damaging the unborn child.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H320	Causes eye irritation.
H402	Harmful to aquatic life.
H290	May be corrosive to metals.
H302	Harmful if swallowed.
H335	May cause respiratory irritation.
H314	Causes severe skin burns and eye damage.
H411	Toxic to aquatic life with long lasting effects.
H318	Causes serious eye damage.
H315	Causes skin irritation.
H317	May cause an allergic skin reaction.

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. The data do not describe the product's properties (product specification). Neither should any agreed property nor the suitability of the product for any specific purpose be deduced from the data contained in the safety data sheet. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.